TensorEngine: informe de corrida

Modelo: `R_plus_alpha_times_RicciUU`
Run ID: `run_a3c369602abdc9f29576`
Estado: `partial`
Backend: `structural-python 0.9.0`
Verificaciones: 43 aprobadas, 0 fallidas, 2 indeterminadas.

Presentación: simplificación protegida, sin modificar los objetos canónicos. Las operaciones e hipótesis usadas por expresión constan en `presentation.json` y en los comentarios del archivo LaTeX. La validación xAct corresponde a la IR canónica.

Deltas canónicos: 1723 sustituciones y 0 trazas registradas en las pasadas del álgebra (incluidas verificaciones); 0 deltas explícitos en las 13 cantidades finales. Detalle e índices sustituidos en `delta_contractions.json`.

Expresiones tensoriales abstractas

Las expresiones de esta sección son covariantes y no incorporan sustituciones del ansatz.

L

$$L = R_{d_0\,d_1\,d_2\,d_3}\, g^{d_0\,d_2}\, g^{d_1\,d_3} + R_{d_0\,d_1\,d_2\,d_3}\, g^{d_0\,d_2}\, g^{d_1\,d_4}\, g^{d_3\,d_5}\, u_{d_4}\, u_{d_5}\, \alpha$$

Significado: Densidad lagrangiana escalar normalizada de la teoría.
Índices libres y varianzas: $\varnothing$ (escalar)
Fuentes del pipeline: `validated_model`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

M_{ab}

$$M_{ab} = -R_{a\,d_0}{}^{d_0\,d_1}\, u_b\, u_{d_1}\, \alpha - u_a\, R_{b\,d_0}{}^{d_0\,d_1}\, u_{d_1}\, \alpha + 2\, R_{a\,d_0\,b}{}^{d_0} + R_a{}^{d_0}{}_b{}^{d_1}\, u_{d_0}\, u_{d_1}\, \alpha$$

Significado: Derivada de L respecto de la métrica inversa, M_ab.
Índices libres y varianzas: $a_\downarrow$, $b_\downarrow$
Fuentes del pipeline: `lagrangian`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

P^{abcd}

$$P^{abcd} = -\frac{g^{a\,d}\, g^{b\,c}}{2} + \frac{g^{a\,c}\, g^{b\,d}}{2} - \frac{g^{a\,d}\, u^b\, u^c\, \alpha}{4} - \frac{u^a\, g^{b\,c}\, u^d\, \alpha}{4} + \frac{g^{a\,c}\, u^b\, u^d\, \alpha}{4} + \frac{u^a\, g^{b\,d}\, u^c\, \alpha}{4}$$

Significado: Momento de curvatura P^abcd=partial L/partial R_abcd.
Índices libres y varianzas: $a^\uparrow$, $b^\uparrow$, $c^\uparrow$, $d^\uparrow$
Fuentes del pipeline: `lagrangian`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

J^a

$$J^a = -2\, R^{a}{}_{d_0}{}^{d_0\, d_1}\, u_{d_1}\, \alpha$$

Significado: Momento J^a conjugado a nabla_a phi.
Índices libres y varianzas: $a^{\uparrow}$
Fuentes del pipeline: `lagrangian`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

F_ϕ

$$F_\phi = 0$$

Significado: Derivada explícita F_phi=partial L/partial phi.
Índices libres y varianzas: $\varnothing$ (escalar)
Fuentes del pipeline: `lagrangian`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

E_{ab}

$$\begin{aligned}
E_{ab} = {} & -R_{a\,d_0}{}^{d_0\,d_1}\, u_b\, u_{d_1}\, \alpha - u_a\, R_{b\,d_0}{}^{d_0\,d_1}\, u_{d_1}\, \alpha - \frac{\nabla_a\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{b\,d_1}\, \alpha}{2} - \frac{\nabla_a\big(u^{d_1}\big)\ \nabla_{d_0}\big(u^{d_0}\big)\ g_{b\,d_1}\, \alpha}{2} \\
& - \frac{\nabla_b\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{a\,d_1}\, \alpha}{2} - \frac{\nabla_b\big(u^{d_1}\big)\ \nabla_{d_0}\big(u^{d_0}\big)\ g_{a\,d_1}\, \alpha}{2} - \frac{g_{a\,b}\, R_{d_0\,d_1}{}^{d_0\,d_1}}{2} \\
& - \frac{g_{a\,b}\, R_{d_0}{}^{d_1\,d_0\,d_2}\, u_{d_1}\, u_{d_2}\, \alpha}{2} - \frac{R_{a\,d_0\,b\,d_1}\, u^{d_0}\, u^{d_1}\, \alpha}{2} + \frac{\nabla_{d_0}\Big(\nabla_{d_1}\big(u^{d_0}\big)\Big)\ g_{a\,b}\, u^{d_1}\, \alpha}{2} \\
& + \frac{\nabla_{d_0}\big(u^{d_0}\big)\ \nabla_{d_1}\big(u^{d_1}\big)\ g_{a\,b}\, \alpha}{2} + \frac{\nabla_{d_1}\Big(\nabla_{d_0}\big(u^{d_0}\big)\Big)\ g_{a\,b}\, u^{d_1}\, \alpha}{2} + \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{a\,b}\, \alpha}{2} \\
& + \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_3}\big(u^{d_2}\big)\ g_{a\,d_0}\, g_{b\,d_2}\, g^{d_1\,d_3}\, \alpha}{2} + \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_3}\big(u^{d_2}\big)\ g_{a\,d_2}\, g_{b\,d_0}\, g^{d_1\,d_3}\, \alpha}{2} \\
& + \frac{\nabla_{d_2}\Big(\nabla_{d_1}\big(u^{d_0}\big)\Big)\ g_{a\,d_0}\, u_b\, g^{d_1\,d_2}\, \alpha}{2} + \frac{\nabla_{d_2}\Big(\nabla_{d_1}\big(u^{d_0}\big)\Big)\ u_a\, g_{b\,d_0}\, g^{d_1\,d_2}\, \alpha}{2} - \frac{\nabla_a\Big(\nabla_{d_0}\big(u^{d_0}\big)\Big)\ u_b\, \alpha}{4} \\
& - \frac{\nabla_a\Big(\nabla_{d_1}\big(u^{d_0}\big)\Big)\ g_{b\,d_0}\, u^{d_1}\, \alpha}{4} - \frac{\nabla_b\Big(\nabla_{d_0}\big(u^{d_0}\big)\Big)\ u_a\, \alpha}{4} - \frac{\nabla_b\Big(\nabla_{d_1}\big(u^{d_0}\big)\Big)\ g_{a\,d_0}\, u^{d_1}\, \alpha}{4} \\
& - \frac{\nabla_{d_0}\Big(\nabla_a\big(u^{d_0}\big)\Big)\ u_b\, \alpha}{4} - \frac{\nabla_{d_0}\Big(\nabla_b\big(u^{d_0}\big)\Big)\ u_a\, \alpha}{4} - \frac{\nabla_{d_1}\Big(\nabla_a\big(u^{d_0}\big)\Big)\ g_{b\,d_0}\, u^{d_1}\, \alpha}{4} \\
& - \frac{\nabla_{d_1}\Big(\nabla_b\big(u^{d_0}\big)\Big)\ g_{a\,d_0}\, u^{d_1}\, \alpha}{4} + \frac{R_{a\,d_0}{}^{d_0}{}_{d_1}\, u_b\, u^{d_1}\, \alpha}{4} + \frac{u_a\, R_{b\,d_0}{}^{d_0}{}_{d_1}\, u^{d_1}\, \alpha}{4} + R_{a}{}^{d_0}{}_{b}{}^{d_1}\, u_{d_0}\, u_{d_1}\, \alpha + R_{a\,d_0\,b}{}^{d_0}
\end{aligned}$$

Significado: Ecuación de Euler-Lagrange métrica E_ab.
Índices libres y varianzas: $a_\downarrow$, $b_\downarrow$
Fuentes del pipeline: `delta_lagrangian, curvature_derivative_metric_term`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$$\begin{aligned}E_{ab}|_{\nabla\nabla P} = &-\frac{\nabla_a\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{b\,d_1}\,\alpha}{2} - \frac{\nabla_a\big(u^{d_1}\big)\ \nabla_{d_0}\big(u^{d_0}\big)\ g_{b\,d_1}\,\alpha}{2}\\
&- \frac{\nabla_b\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{a\,d_1}\,\alpha}{2} - \frac{\nabla_b\big(u^{d_1}\big)\ \nabla_{d_0}\big(u^{d_0}\big)\ g_{a\,d_1}\,\alpha}{2}\\
&+ \frac{\nabla_{d_0}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ g_{a\,b}\,u^{d_1}\,\alpha}{2} + \frac{\nabla_{d_0}\big(u^{d_0}\big)\ \nabla_{d_1}\big(u^{d_1}\big)\ g_{a\,b}\,\alpha}{2} + \frac{\nabla_{d_1}\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\ g_{a\,b}\,u^{d_1}\,\alpha}{2}\\
&+ \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{a\,b}\,\alpha}{2} + \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_3}\big(u^{d_2}\big)\ g_{a\,d_0}\,g_{b\,d_2}\,g^{d_1\,d_3}\,\alpha}{2}\\
&+ \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_3}\big(u^{d_2}\big)\ g_{a\,d_2}\,g_{b\,d_0}\,g^{d_1\,d_3}\,\alpha}{2} + \frac{\nabla_{d_2}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ g_{a\,d_0}\,u_b\,g^{d_1\,d_2}\,\alpha}{2}\\
&+ \frac{\nabla_{d_2}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ u_a\,g_{b\,d_0}\,g^{d_1\,d_2}\,\alpha}{2} - \frac{\nabla_a\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\ u_b\,\alpha}{4} - \frac{\nabla_a\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ g_{b\,d_0}\,u^{d_1}\,\alpha}{4}\\
&- \frac{\nabla_b\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\ u_a\,\alpha}{4} - \frac{\nabla_b\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ g_{a\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_{d_0}\big(\nabla_a\big(u^{d_0}\big)\big)\ u_b\,\alpha}{4}\\
&- \frac{\nabla_{d_0}\big(\nabla_b\big(u^{d_0}\big)\big)\ u_a\,\alpha}{4} - \frac{\nabla_{d_1}\big(\nabla_a\big(u^{d_0}\big)\big)\ g_{b\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_{d_1}\big(\nabla_b\big(u^{d_0}\big)\big)\ g_{a\,d_0}\,u^{d_1}\,\alpha}{4}\end{aligned}$$

La expresión anterior identifica la contribución $-2\nabla^c\nabla^d P_{acdb}$ dentro de E_{ab}.

E_ϕ

$$E_\phi = -2\,\nabla_{d_2}(u_{d_1})\ R_{d_0}{}^{d_2\,d_0\,d_1}\,\alpha + 2\,\nabla_{d_1}\Big(-R_{d_0}{}^{d_1\,d_0\,d_2}\Big)\ u_{d_2}\,\alpha$$

Significado: Ecuación de Euler-Lagrange escalar E_phi.
Índices libres y varianzas: $\varnothing$ (escalar)
Fuentes del pipeline: `scalar_derivative, scalar_gradient_momentum`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$\mathcal{R}$

$$\mathcal{R} = g^{a\,c}\,g^{b\,d}\,R_{a\,b\,c\,d}$$

Significado: Escalar de Ricci construido con la convención geométrica activa.
Índices libres y varianzas: $\varnothing$ (escalar)
Fuentes del pipeline: `validated_model`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$R_{ab}R^{ab}$

$$R_{ab}R^{ab} = g^{a\,c}\,R_{a\,b\,c\,d}\,g^{b\,e}\,g^{d\,f}\,g^{g\,h}\,R_{g\,e\,h\,f}$$

Significado: Invariante cuadrático de Ricci R_ab R^ab.
Índices libres y varianzas: $\varnothing$ (escalar)
Fuentes del pipeline: `validated_model, riemann_tensor`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

R_{abcd}

$$R_{abcd} = R_{a\,b\,c\,d}$$

Significado: Tensor de Riemann covariante tratado como entrada geométrica.
Índices libres y varianzas: $a_{\downarrow}$, $b_{\downarrow}$, $c_{\downarrow}$, $d_{\downarrow}$
Fuentes del pipeline: `validated_model`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$R_{abcd}R^{abcd}$

$$R_{abcd}R^{abcd} = g^{a\,e}\,g^{b\,f}\,g^{c\,g}\,g^{d\,h}\,R_{a\,b\,c\,d}\,R_{e\,f\,g\,h}$$

Significado: Invariante de Kretschmann R_abcd R^abcd.
Índices libres y varianzas: $\varnothing$ (escalar)
Fuentes del pipeline: `validated_model, riemann_tensor`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$\nabla_e P^{abcd}$

$$\nabla_e P^{abcd} = \frac{1}{4}\,\alpha\,\Big(\nabla_e(u^a)\;g^{b\,d}\,u^c + \nabla_e\Big(u^b\Big)\;g^{a\,c}\,u^d + \nabla_e(u^c)\;u^a\,g^{b\,d} + \nabla_e\Big(u^d\Big)\;g^{a\,c}\,u^b - \nabla_e(u^a)\;g^{b\,c}\,u^d - \nabla_e\Big(u^b\Big)\;g^{a\,d}\,u^c - \nabla_e(u^c)\;g^{a\,d}\,u^b - \nabla_e\Big(u^d\Big)\;u^a\,g^{b\,c}\Big)$$

Significado: Primera derivada covariante del momento de curvatura.
Índices libres y varianzas: $a^{\uparrow}$, $b^{\uparrow}$, $c^{\uparrow}$, $d^{\uparrow}$, $e_{\downarrow}$
Fuentes del pipeline: `curvature_momentum`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$\nabla_f \nabla_e P^{abcd}$

$$\begin{aligned}\nabla_f \nabla_e P^{abcd} = \frac{1}{4}\,\alpha\,\Big(&\nabla_e(u^a)\ \nabla_f(u^c)\ g^{b\,d} + \nabla_e\Big(u^b\Big)\ \nabla_f\Big(u^d\Big)\ g^{a\,c} + \nabla_e(u^c)\ \nabla_f(u^a)\ g^{b\,d} \\ &+ \nabla_e\Big(u^d\Big)\ \nabla_f\Big(u^b\Big)\ g^{a\,c} + \nabla_f(\nabla_e(u^a))\ g^{b\,d}\,u^c + \nabla_f\Big(\nabla_e\Big(u^b\Big)\Big)\ g^{a\,c}\,u^d \\ &+ \nabla_f(\nabla_e(u^c))\ u^a\,g^{b\,d} + \nabla_f\Big(\nabla_e\Big(u^d\Big)\Big)\ g^{a\,c}\,u^b - \nabla_e(u^a)\ \nabla_f\Big(u^d\Big)\ g^{b\,c} \\ &- \nabla_e\Big(u^b\Big)\ \nabla_f(u^c)\ g^{a\,d} - \nabla_e(u^c)\ \nabla_f\Big(u^b\Big)\ g^{a\,d} - \nabla_e\Big(u^d\Big)\ \nabla_f(u^a)\ g^{b\,c} - \nabla_f(\nabla_e(u^a))\ g^{b\,c}\,u^d \\ &- \nabla_f\Big(\nabla_e\Big(u^b\Big)\Big)\ g^{a\,d}\,u^c - \nabla_f(\nabla_e(u^c))\ g^{a\,d}\,u^b - \nabla_f\Big(\nabla_e\Big(u^d\Big)\Big)\ u^a\,g^{b\,c}\Big)\end{aligned}$$

Significado: Segunda derivada covariante del momento de curvatura.
Índices libres y varianzas: $a^{\uparrow}$, $b^{\uparrow}$, $c^{\uparrow}$, $d^{\uparrow}$, $e_{\downarrow}$, $f_{\downarrow}$
Fuentes del pipeline: `nabla_P`
Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

Descomposición compacta adicional

Esta vista se agrega después de los resultados anteriores y reutiliza exactamente la IR canónica ya calculada. No interviene en la validación ni en los fingerprints.

Descomposición de E_{ab}.

$$E_{ab} = M_{ab} - \mathcal{R}_{(ab)}[P] - \frac{1}{2} g_{ab} L - 2\nabla_m \nabla_n P^{mn}{}_{ab}$$

Reconstrucción IR: verified. Los cuatro bloques reconstruyen exactamente E_ab en la IR canónica.

forma compacta:
$$\begin{aligned}E_{ab} = &-R_{a\,d_0}{}^{d_0\,d_1}\,u_b\,u_{d_1}\,\alpha - u_a\,R_{b\,d_0}{}^{d_0\,d_1}\,u_{d_1}\,\alpha - \frac{\nabla_a\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{b\,d_1}\,\alpha}{2} \\ &- \frac{\nabla_a\big(u^{d_1}\big)\ \nabla_{d_0}\big(u^{d_0}\big)\ g_{b\,d_1}\,\alpha}{2} - \frac{\nabla_b\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{a\,d_1}\,\alpha}{2} \\ &- \frac{\nabla_b\big(u^{d_1}\big)\ \nabla_{d_0}\big(u^{d_0}\big)\ g_{a\,d_1}\,\alpha}{2} - \frac{g_{a\,b}\,R_{d_0\,d_1}{}^{d_0\,d_1}}{2} - \frac{g_{a\,b}\,R_{d_0}{}^{d_1\,d_0\,d_2}\,u_{d_1}\,u_{d_2}\,\alpha}{2} \\ &- \frac{R_{a\,d_0\,b\,d_1}\,u^{d_0}\,u^{d_1}\,\alpha}{2} + \frac{\nabla_{d_0}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ g_{a\,b}\,u^{d_1}\,\alpha}{2} + \frac{\nabla_{d_0}\big(u^{d_0}\big)\ \nabla_{d_1}\big(u^{d_1}\big)\ g_{a\,b}\,\alpha}{2} \\ &+ \frac{\nabla_{d_1}\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\ g_{a\,b}\,u^{d_1}\,\alpha}{2} + \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_0}\big(u^{d_1}\big)\ g_{a\,b}\,\alpha}{2} \\ &+ \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_3}\big(u^{d_2}\big)\ g_{a\,d_0}\,g_{b\,d_2}\,g^{d_1\,d_3}\,\alpha}{2} + \frac{\nabla_{d_1}\big(u^{d_0}\big)\ \nabla_{d_3}\big(u^{d_2}\big)\ g_{a\,d_2}\,g_{b\,d_0}\,g^{d_1\,d_3}\,\alpha}{2} \\ &+ \frac{\nabla_{d_2}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ g_{a\,d_0}\,u_b\,g^{d_1\,d_2}\,\alpha}{2} + \frac{\nabla_{d_2}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ u_a\,g_{b\,d_0}\,g^{d_1\,d_2}\,\alpha}{2} \\ &- \frac{\nabla_a\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\ u_b\,\alpha}{4} - \frac{\nabla_a\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ g_{b\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_b\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\ u_a\,\alpha}{4} \\ &- \frac{\nabla_b\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\ g_{a\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_{d_0}\big(\nabla_a\big(u^{d_0}\big)\big)\ u_b\,\alpha}{4} - \frac{\nabla_{d_0}\big(\nabla_b\big(u^{d_0}\big)\big)\ u_a\,\alpha}{4} \\ &- \frac{\nabla_{d_1}\big(\nabla_a\big(u^{d_0}\big)\big)\ g_{b\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_{d_1}\big(\nabla_b\big(u^{d_0}\big)\big)\ g_{a\,d_0}\,u^{d_1}\,\alpha}{4} \\ &+ \frac{R_{a\,d_0}{}^{d_0}{}_{d_1}\,u_b\,u^{d_1}\,\alpha}{4} + \frac{u_a\,R_{b\,d_0}{}^{d_0}{}_{d_1}\,u^{d_1}\,\alpha}{4} + R_a{}^{d_0}{}_b{}^{d_1}\,u_{d_0}\,u_{d_1}\,\alpha + R_{a\,d_0\,b}{}^{d_0}\end{aligned}$$

Bloque M_{ab} (fuentes: `metric_momentum`).

forma compacta: $M_{ab} = -R_{a\,d_0}{}^{d_0\,d_1}\,u_b\,u_{d_1}\,\alpha - u_a\,R_{b\,d_0}{}^{d_0\,d_1}\,u_{d_1}\,\alpha + 2\,R_{a\,d_0\,b}{}^{d_0} + R_a{}^{d_0}{}_b{}^{d_1}\,u_{d_0}\,u_{d_1}\,\alpha$

Bloque $-\mathcal{R}_{(ab)}[P]$ (fuentes: `curvature_momentum, riemann_tensor`).

forma compacta:
$$
\begin{aligned}
-\mathcal{R}_{(ab)}[P] = \frac{1}{2}\Bigg(&-g_{a\,d_0}\left(-\frac{g^{d_0\,d_1}\,g^{d_2\,d_3}}{2} + \frac{g^{d_0\,d_3}\,g^{d_2\,d_1}}{2} - \frac{g^{d_0\,d_1}\,u^{d_2}\,u^{d_3}\,\alpha}{4}\right.\\
&\left. - \frac{u^{d_0}\,g^{d_2\,d_3}\,u^{d_1}\,\alpha}{4} + \frac{g^{d_0\,d_3}\,u^{d_2}\,u^{d_1}\,\alpha}{4} + \frac{u^{d_0}\,g^{d_2\,d_1}\,u^{d_3}\,\alpha}{4}\right) R_{b\,d_2\,d_3\,d_1}\\
&- g_{b\,d_0}\left(-\frac{g^{d_0\,d_1}\,g^{d_2\,d_3}}{2} + \frac{g^{d_0\,d_3}\,g^{d_2\,d_1}}{2} - \frac{g^{d_0\,d_1}\,u^{d_2}\,u^{d_3}\,\alpha}{4}\right.\\
&\left. - \frac{u^{d_0}\,g^{d_2\,d_3}\,u^{d_1}\,\alpha}{4} + \frac{g^{d_0\,d_3}\,u^{d_2}\,u^{d_1}\,\alpha}{4} + \frac{u^{d_0}\,g^{d_2\,d_1}\,u^{d_3}\,\alpha}{4}\right) R_{a\,d_2\,d_3\,d_1}\Bigg)
\end{aligned}
$$

Bloque $-\frac{1}{2}g_{ab}L$ (fuentes: `lagrangian`).

forma compacta: $-\frac{1}{2}g_{ab}L = -\frac{1}{2}\,g_{a\,b}\left(R_{d_0\,d_1\,d_2\,d_3}\,g^{d_0\,d_2}\,g^{d_1\,d_3} + R_{d_0\,d_1\,d_2\,d_3}\,g^{d_0\,d_2}\,g^{d_1\,d_4}\,g^{d_3\,d_5}\,u_{d_4}\,u_{d_5}\,\alpha\right)$

Bloque $E_{ab}^{(\nabla\nabla P)}$ (fuentes: `curvature_momentum, nabla_nabla_P`).

forma compacta:
$$
\begin{aligned}
E_{ab}^{(\nabla\nabla P)} = &-\frac{\nabla_a\left(u^{d_0}\right)\,\nabla_{d_0}\left(u^{d_1}\right)\,g_{b\,d_1}\,\alpha}{2} - \frac{\nabla_a\left(u^{d_1}\right)\,\nabla_{d_0}\left(u^{d_0}\right)\,g_{b\,d_1}\,\alpha}{2}\\
&- \frac{\nabla_b\left(u^{d_0}\right)\,\nabla_{d_0}\left(u^{d_1}\right)\,g_{a\,d_1}\,\alpha}{2} - \frac{\nabla_b\left(u^{d_1}\right)\,\nabla_{d_0}\left(u^{d_0}\right)\,g_{a\,d_1}\,\alpha}{2}\\
&+ \frac{\nabla_{d_0}\left(\nabla_{d_1}\left(u^{d_0}\right)\right)\,g_{a\,b}\,u^{d_1}\,\alpha}{2} + \frac{\nabla_{d_0}\left(u^{d_0}\right)\,\nabla_{d_1}\left(u^{d_1}\right)\,g_{a\,b}\,\alpha}{2}\\
&+ \frac{\nabla_{d_1}\left(\nabla_{d_0}\left(u^{d_0}\right)\right)\,g_{a\,b}\,u^{d_1}\,\alpha}{2} + \frac{\nabla_{d_1}\left(u^{d_0}\right)\,\nabla_{d_0}\left(u^{d_1}\right)\,g_{a\,b}\,\alpha}{2}\\
&+ \frac{\nabla_{d_1}\left(u^{d_0}\right)\,\nabla_{d_3}\left(u^{d_2}\right)\,g_{a\,d_0}\,g_{b\,d_2}\,g^{d_1\,d_3}\,\alpha}{2}\\
&+ \frac{\nabla_{d_1}\left(u^{d_0}\right)\,\nabla_{d_3}\left(u^{d_2}\right)\,g_{a\,d_2}\,g_{b\,d_0}\,g^{d_1\,d_3}\,\alpha}{2} + \frac{\nabla_{d_2}\left(\nabla_{d_1}\left(u^{d_0}\right)\right)\,g_{a\,d_0}\,u_b\,g^{d_1\,d_2}\,\alpha}{2}\\
&+ \frac{\nabla_{d_2}\left(\nabla_{d_1}\left(u^{d_0}\right)\right)\,u_a\,g_{b\,d_0}\,g^{d_1\,d_2}\,\alpha}{2} - \frac{\nabla_a\left(\nabla_{d_0}\left(u^{d_0}\right)\right)\,u_b\,\alpha}{4}\\
&- \frac{\nabla_a\left(\nabla_{d_1}\left(u^{d_0}\right)\right)\,g_{b\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_b\left(\nabla_{d_0}\left(u^{d_0}\right)\right)\,u_a\,\alpha}{4}\\
&- \frac{\nabla_b\left(\nabla_{d_1}\left(u^{d_0}\right)\right)\,g_{a\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_{d_0}\left(\nabla_a\left(u^{d_0}\right)\right)\,u_b\,\alpha}{4} - \frac{\nabla_{d_0}\left(\nabla_b\left(u^{d_0}\right)\right)\,u_a\,\alpha}{4}\\
&- \frac{\nabla_{d_1}\left(\nabla_a\left(u^{d_0}\right)\right)\,g_{b\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_{d_1}\left(\nabla_b\left(u^{d_0}\right)\right)\,g_{a\,d_0}\,u^{d_1}\,\alpha}{4}
\end{aligned}
$$

Descomposición de E_ϕ.

$$E_\phi = F_\phi - \nabla_a J^a$$

Reconstrucción IR: verified. Los dos bloques reconstruyen exactamente E_phi en la IR canónica.

$$\text{forma compacta:}\quad E_\phi = -2\,\nabla_{d_2}(u_{d_1})\ R_{d_0}{}^{d_2\,d_0\,d_1}\,\alpha + 2\,\nabla_{d_1}\Big(-R_{d_0}{}^{d_1\,d_0\,d_2}\Big)\ u_{d_2}\,\alpha$$

Bloque F_ϕ (fuentes: `scalar_derivative`).

$$\text{forma compacta:}\quad F_\phi = 0$$

Bloque $-\nabla_a J^a$ (fuentes: `scalar_gradient_momentum, scalar_euler`).

$$\text{forma compacta:}\quad -\nabla_a J^a = -2\,\nabla_{d_2}(u_{d_1})\ R_{d_0}{}^{d_2\,d_0\,d_1}\,\alpha + 2\,\nabla_{d_1}\Big(-R_{d_0}{}^{d_1\,d_0\,d_2}\Big)\ u_{d_2}\,\alpha$$

Descomposición de P^{abcd}.

$$P^{abcd} = \partial L/\partial R_{abcd}$$

Reconstrucción IR: verified. La forma compacta y la expandida referencian el mismo objeto P^abcd.

$$\text{forma compacta:}\quad P^{abcd} = -\frac{g^{a\,d}\,g^{b\,c}}{2} + \frac{g^{a\,c}\,g^{b\,d}}{2} - \frac{g^{a\,d}\,u^b\,u^c\,\alpha}{4} - \frac{u^a\,g^{b\,c}\,u^d\,\alpha}{4} + \frac{g^{a\,c}\,u^b\,u^d\,\alpha}{4} + \frac{u^a\,g^{b\,d}\,u^c\,\alpha}{4}$$

Expresiones proyectadas mediante el ansatz

Ansatz utilizado: `draft4_circular`.

$$(x^\mu) = (\tau, r, \varphi)\,, \qquad ds^2 = -f(r)\ d\tau^2 + \frac{1}{f(r)}\,dr^2 + r^2\,d\varphi^2$$

$$\phi = \Phi(r,\varphi) \qquad \text{(genérico, sin especialización)}$$

Las componentes completas se conservan de forma dispersa en `results.json`; el documento muestra como máximo doce componentes no nulas por tensor.

L

Ansatz: `draft4_circular`; **estado:** completada.

$$L|_{\text{ansatz}} = -\frac{1}{2\,r^3}\left(r^2\left(\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2\,f(r) + 4\right)f'(r) + r^3\left(\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2\,f(r) + 2\right)f''(r) + 2\,\alpha\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)^2\,f'(r)\right)$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'.

M_{ab}

Ansatz: `draft4_circular`; **estado:** completada.

$$[M_{ab}]_{00} = \frac{1}{2\,r^3}\,\left(\alpha\,\left(\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)^2\,f'(r) + r^3\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2\,f''(r)\,f(r)\right) + 2\,r^2\,\left(r\,f''(r) + f'(r)\right)\right)\,f(r)$$

$$[M_{ab}]_{11} = -\frac{1}{2\,r^3\,f(r)}\,\left(2\,r^2\,\left(r\,f''(r) + f'(r)\right) + \alpha\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)^2\,f'(r) \right.$$
$$\left. + 2\,\alpha\,r^2\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2\,\left(r\,f''(r) + f'(r)\right)\,f(r)\right)$$

$$[M_{ab}]_{12} = -\frac{1}{2\,r}\,\alpha\,\left(2\,f'(r) + r\,f''(r)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)$$

$$[M_{ab}]_{21} = -\frac{1}{2\,r}\,\alpha\,\left(2\,f'(r) + r\,f''(r)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)$$

$$[M_{ab}]_{22} = -\frac{1}{2\,r}\,\left(r^2\,\left(\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2\,f(r) + 4\right) + 4\,\alpha\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)^2\right)\,f'(r)$$

Proyección completa almacenada: 5 componentes no nulas de 9. Proyectada respetando el ansatz 'draft4_circular'.

P^{abcd}

Ansatz: `draft4_circular`; **estado:** completada.

$$\left[P^{abcd}\right]^{0110} = \frac{\alpha\,f(r)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2}{4} + \frac{1}{2}$$

$$\left[P^{abcd}\right]^{0120} = \frac{\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)}{4\,r^2}$$

$$\left[P^{abcd}\right]^{0210} = \frac{\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)}{4\,r^2}$$

$$\left[P^{abcd}\right]^{0220} = \frac{\left(2\,r^2 + \alpha\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)^2\right)}{4\,r^4\,f(r)}$$

$$\left[P^{abcd}\right]^{0101} = -\frac{\alpha\,f(r)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2}{4} - \frac{1}{2}$$

$$\left[P^{abcd}\right]^{0201} = -\frac{\alpha \frac{\partial^1 \Phi}{\partial r^1}(r,\varphi)\ \frac{\partial^1 \Phi}{\partial \varphi^1}(r,\varphi)}{4\,r^2}$$

$$\left[P^{abcd}\right]^{0102} = -\frac{\alpha \frac{\partial^1 \Phi}{\partial r^1}(r,\varphi)\ \frac{\partial^1 \Phi}{\partial \varphi^1}(r,\varphi)}{4\,r^2}$$

$$\left[P^{abcd}\right]^{0202} = -\frac{\left(2\,r^2 + \alpha \frac{\partial^1 \Phi}{\partial \varphi^1}(r,\varphi)^2\right)}{4\,r^4\,f(r)}$$

$$\left[P^{abcd}\right]^{1010} = -\frac{\alpha\,f(r)\ \frac{\partial^1 \Phi}{\partial r^1}(r,\varphi)^2}{4} - \frac{1}{2}$$

$$\left[P^{abcd}\right]^{1020} = -\frac{\alpha \frac{\partial^1 \Phi}{\partial r^1}(r,\varphi)\ \frac{\partial^1 \Phi}{\partial \varphi^1}(r,\varphi)}{4\,r^2}$$

$$\left[P^{abcd}\right]^{1001} = \frac{\alpha\,f(r)\ \frac{\partial^1 \Phi}{\partial r^1}(r,\varphi)^2}{4} + \frac{1}{2}$$

$$\left[P^{abcd}\right]^{1221} = -\frac{1}{4\,r^4}\left(\alpha \frac{\partial^1 \Phi}{\partial \varphi^1}(r,\varphi)^2 + r^2\left(\alpha \frac{\partial^1 \Phi}{\partial r^1}(r,\varphi)^2 f(r) + 2\right)\right) f(r)$$

Representación compacta: se muestran 12 de 20 componentes no nulas. Proyección completa almacenada: 20 componentes no nulas de 81. Proyectada respetando el ansatz 'draft4_circular'.

J^a

Ansatz: `draft4_circular`; **estado:** completada.

$$[J^a]^1 = -\frac{\alpha\ (r\,f''(r) + f'(r))\ \frac{\partial^1 \Phi}{\partial r^1}(r,\varphi)\ f(r)}{r}$$

$$[J^a]^2 = -\frac{2\,\alpha \frac{\partial^1 \Phi}{\partial \varphi^1}(r,\varphi)\ f'(r)}{r^3}$$

Proyección completa almacenada: 2 componentes no nulas de 3. Proyectada respetando el ansatz 'draft4_circular'.

F_ϕ

Ansatz: `draft4_circular`; **estado:** completada.

$$F_\phi|_{\mathrm{ansatz}} = 0$$

Proyección completa almacenada: 0 componentes no nulas de 1. La expresión abstracta es nula; todas sus componentes son cero.

E_{ab}

Ansatz: `draft4_circular`; **estado:** no disponible por limitación del backend.

La forma abstracta se conserva sin sustitución:

$$\begin{aligned}
E_{ab} = &-R_{a\,d_0}{}^{d_0\,d_1}\,u_b\,u_{d_1}\,\alpha - u_a\,R_{b\,d_0}{}^{d_0\,d_1}\,u_{d_1}\,\alpha - \frac{\nabla_a\big(u^{d_0}\big)\;\nabla_{d_0}\big(u^{d_1}\big)\;g_{b\,d_1}\,\alpha}{2} - \frac{\nabla_a\big(u^{d_1}\big)\;\nabla_{d_0}\big(u^{d_0}\big)\;g_{b\,d_1}\,\alpha}{2} \\
&- \frac{\nabla_b\big(u^{d_0}\big)\;\nabla_{d_0}\big(u^{d_1}\big)\;g_{a\,d_1}\,\alpha}{2} - \frac{\nabla_b\big(u^{d_1}\big)\;\nabla_{d_0}\big(u^{d_0}\big)\;g_{a\,d_1}\,\alpha}{2} - \frac{g_{a\,b}\,R_{d_0\,d_1}{}^{d_0\,d_1}}{2} \\
&- \frac{g_{a\,b}\,R_{d_0}{}^{d_1\,d_0\,d_2}\,u_{d_1}\,u_{d_2}\,\alpha}{2} - \frac{R_{a\,d_0\,b\,d_1}\,u^{d_0}\,u^{d_1}\,\alpha}{2} + \frac{\nabla_{d_0}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\;g_{a\,b}\,u^{d_1}\,\alpha}{2} \\
&+ \frac{\nabla_{d_0}\big(u^{d_0}\big)\;\nabla_{d_1}\big(u^{d_1}\big)\;g_{a\,b}\,\alpha}{2} + \frac{\nabla_{d_1}\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\;g_{a\,b}\,u^{d_1}\,\alpha}{2} + \frac{\nabla_{d_1}\big(u^{d_0}\big)\;\nabla_{d_0}\big(u^{d_1}\big)\;g_{a\,b}\,\alpha}{2} \\
&+ \frac{\nabla_{d_1}\big(u^{d_0}\big)\;\nabla_{d_3}\big(u^{d_2}\big)\;g_{a\,d_0}\,g_{b\,d_2}\,g^{d_1\,d_3}\,\alpha}{2} + \frac{\nabla_{d_1}\big(u^{d_0}\big)\;\nabla_{d_3}\big(u^{d_2}\big)\;g_{a\,d_2}\,g_{b\,d_0}\,g^{d_1\,d_3}\,\alpha}{2} \\
&+ \frac{\nabla_{d_2}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\;g_{a\,d_0}\,u_b\,g^{d_1\,d_2}\,\alpha}{2} + \frac{\nabla_{d_2}\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\;u_a\,g_{b\,d_0}\,g^{d_1\,d_2}\,\alpha}{2} - \frac{\nabla_a\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\;u_b\,\alpha}{4} \\
&- \frac{\nabla_a\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\;g_{b\,d_0}\,u^{d_1}\,\alpha}{4} - \frac{\nabla_b\big(\nabla_{d_0}\big(u^{d_0}\big)\big)\;u_a\,\alpha}{4} - \frac{\nabla_b\big(\nabla_{d_1}\big(u^{d_0}\big)\big)\;g_{a\,d_0}\,u^{d_1}\,\alpha}{4} \\
&- \frac{\nabla_{d_0}\big(\nabla_a\big(u^{d_0}\big)\big)\;u_b\,\alpha}{4} - \frac{\nabla_{d_0}\big(\nabla_b\big(u^{d_0}\big)\big)\;u_a\,\alpha}{4} - \frac{\nabla_{d_1}\big(\nabla_a\big(u^{d_0}\big)\big)\;g_{b\,d_0}\,u^{d_1}\,\alpha}{4} \\
&- \frac{\nabla_{d_1}\big(\nabla_b\big(u^{d_0}\big)\big)\;g_{a\,d_0}\,u^{d_1}\,\alpha}{4} + \frac{R_{a\,d_0}{}^{d_0}{}_{d_1}\,u_b\,u^{d_1}\,\alpha}{4} + \frac{u_a\,R_{b\,d_0}{}^{d_0}{}_{d_1}\,u^{d_1}\,\alpha}{4} + R_a{}^{d_0}{}_b{}^{d_1}\,u_{d_0}\,u_{d_1}\,\alpha + R_{a\,d_0\,b}{}^{d_0}
\end{aligned}$$

Motivo: La expresión se conserva simbólica: su proyección con el campo genérico Phi(r, varphi) contiene 40 nodos de derivada covariante, por encima del límite seguro 24 del backend. Especialice el perfil escalar o eleve el límite del backend de componentes.

E_ϕ

Ansatz: `draft4_circular`; **estado:** completada.

$$\begin{aligned}
E_\phi|_{\mathrm{ansatz}} = \frac{1}{r^3}\,\alpha\,\Bigg(2\,\left(r\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\;f(r) + \frac{\partial^2\Phi}{\partial\varphi^2}(r,\varphi)\right)\,f'(r) \\
+ r^2\,\left(r\,f''(r) + f'(r)\right)\,\left(\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\;f'(r) + \frac{\partial^2\Phi}{\partial r^2}(r,\varphi)\;f(r)\right) \\
+ r\,\left(-2\,f'(r) + r\,\left(2\,f''(r) + r\,f^{(3)}(r)\right)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\;f(r)\Bigg)
\end{aligned}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'.

$\mathcal{R}$

Ansatz: `draft4_circular`; **estado:** completada.

$$\mathcal{R}|_{\text{ansatz}} = -\left(\frac{2\,f'(r)}{r} + f''(r)\right)$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$R_{ab}R^{ab}$

Ansatz: `draft4_circular`; **estado:** completada.

$$R_{ab}R^{ab}|_{\text{ansatz}} = \frac{f''(r)^2}{2} + \frac{3\,f'(r)^2}{2\,r^2} + \frac{f'(r)\;f''(r)}{r}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

R_{abcd}

Ansatz: `draft4_circular`; **estado:** completada.

$$[R_{abcd}]_{0101} = \frac{f''(r)}{2}$$

$$[R_{abcd}]_{0110} = -\frac{f''(r)}{2}$$

$$[R_{abcd}]_{0202} = \frac{r\,f'(r)\;f(r)}{2}$$

$$[R_{abcd}]_{0220} = -\frac{r\,f'(r)\;f(r)}{2}$$

$$[R_{abcd}]_{1001} = -\frac{f''(r)}{2}$$

$$[R_{abcd}]_{1010} = \frac{f''(r)}{2}$$

$$[R_{abcd}]_{1212} = -\frac{r\,f'(r)}{2\,f(r)}$$

$$[R_{abcd}]_{1221} = \frac{r\,f'(r)}{2\,f(r)}$$

$$[R_{abcd}]_{2002} = -\frac{r\,f'(r)\;f(r)}{2}$$

$$[R_{abcd}]_{2020} = \frac{r\,f'(r)\;f(r)}{2}$$

$$[R_{abcd}]_{2112} = \frac{r\,f'(r)}{2\,f(r)}$$

$$[R_{abcd}]_{2121} = -\frac{r\,f'(r)}{2\,f(r)}$$

Proyección completa almacenada: 12 componentes no nulas de 81. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$R_{abcd}R^{abcd}$

Ansatz: `draft4_circular`; **estado:** completada.

$$R_{abcd}R^{abcd}|_{\text{ansatz}} = f''(r)^2 + \frac{2\,f'(r)^2}{r^2}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$\nabla_e P^{abcd}$

Ansatz: `draft4_circular`; **estado:** completada.

$$\left[\nabla_e P^{abcd}\right]_0^{0112} = -\frac{1}{8\,r^2}\,\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\;\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)\;f'(r)\;f(r)$$

$$\left[\nabla_e P^{abcd}\right]_0^{0121} = \frac{1}{8\,r^2}\,\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\;\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)\;f'(r)\;f(r)$$

$$\left[\nabla_e P^{abcd}\right]_0^{0212} = \frac{\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2\;f'(r)\;f(r)}{8\,r^2}$$

$$\left[\nabla_e P^{abcd}\right]_0^{0221} = -\frac{\alpha\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)^2\;f'(r)\;f(r)}{8\,r^2}$$

$$\left[\nabla_e P^{abcd}\right]_1^{0101} = -\frac{1}{4}\,\alpha\,\left(\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\;f'(r) + 2\,\frac{\partial^2\Phi}{\partial r^2}(r,\varphi)\;f(r)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)$$

$$\left[\nabla_e P^{abcd}\right]_1^{0102} = -\frac{1}{8\,r^3\,f(r)}\,\alpha\,\left(r\,\left(\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f'(r) + 2\,\frac{\partial^2\Phi}{\partial r^2}(r,\varphi)\,f(r)\right)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + 2\,\left(-\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + r\,\frac{\partial^2\Phi}{\partial r^1\partial\varphi^1}(r,\varphi)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f(r)\right)$$

$$\left[\nabla_e P^{abcd}\right]_1^{0110} = \frac{1}{4}\,\alpha\,\left(\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f'(r) + 2\,\frac{\partial^2\Phi}{\partial r^2}(r,\varphi)\,f(r)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)$$

$$\left[\nabla_e P^{abcd}\right]_1^{0120} = \frac{1}{8\,r^3\,f(r)}\,\alpha\,\left(r\,\left(\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f'(r) + 2\,\frac{\partial^2\Phi}{\partial r^2}(r,\varphi)\,f(r)\right)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + 2\,\left(-\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + r\,\frac{\partial^2\Phi}{\partial r^1\partial\varphi^1}(r,\varphi)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f(r)\right)$$

$$\left[\nabla_e P^{abcd}\right]_1^{0201} = -\frac{1}{8\,r^3\,f(r)}\,\alpha\,\left(r\,\left(\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f'(r) + 2\,\frac{\partial^2\Phi}{\partial r^2}(r,\varphi)\,f(r)\right)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + 2\,\left(-\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + r\,\frac{\partial^2\Phi}{\partial r^1\partial\varphi^1}(r,\varphi)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f(r)\right)$$

$$\left[\nabla_e P^{abcd}\right]_1^{0202} = \frac{1}{2\,r^5\,f(r)}\,\alpha\,\left(-r\,\frac{\partial^2\Phi}{\partial r^1\partial\varphi^1}(r,\varphi) + \frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)\right)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)$$

$$\left[\nabla_e P^{abcd}\right]_1^{0210} = \frac{1}{8\,r^3\,f(r)}\,\alpha\,\left(r\,\left(\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f'(r) + 2\,\frac{\partial^2\Phi}{\partial r^2}(r,\varphi)\,f(r)\right)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + 2\,\left(-\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + r\,\frac{\partial^2\Phi}{\partial r^1\partial\varphi^1}(r,\varphi)\right)\,\frac{\partial^1\Phi}{\partial r^1}(r,\varphi)\,f(r)\right)$$

$$\left[\nabla_e P^{abcd}\right]_1^{0220} = \frac{1}{2\,r^5\,f(r)}\,\alpha\,\left(-\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi) + r\,\frac{\partial^2\Phi}{\partial r^1\partial\varphi^1}(r,\varphi)\right)\,\frac{\partial^1\Phi}{\partial\varphi^1}(r,\varphi)$$

Representación compacta: se muestran 12 de 56 componentes no nulas. Proyección completa almacenada: 56 componentes no nulas de 243. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$\nabla_f \nabla_e P^{abcd}$

Ansatz: `draft4_circular`; **estado:** no disponible por limitación del backend.

La forma abstracta se conserva sin sustitución:

$$\begin{aligned}\nabla_f\nabla_e P^{abcd} = \frac{1}{4}\,\alpha\,\Big(&\nabla_e(u^a)\,\nabla_f(u^c)\,g^{bd} + \nabla_e\left(u^b\right)\,\nabla_f\left(u^d\right)\,g^{ac} + \nabla_e(u^c)\,\nabla_f(u^a)\,g^{bd} \\ &+ \nabla_e\left(u^d\right)\,\nabla_f\left(u^b\right)\,g^{ac} + \nabla_f(\nabla_e(u^a))\,g^{bd}\,u^c + \nabla_f\left(\nabla_e\left(u^b\right)\right)\,g^{ac}\,u^d \\ &+ \nabla_f(\nabla_e(u^c))\,u^a\,g^{bd} + \nabla_f\left(\nabla_e\left(u^d\right)\right)\,g^{ac}\,u^b - \nabla_e(u^a)\,\nabla_f\left(u^d\right)\,g^{bc} \\ &- \nabla_e\left(u^b\right)\,\nabla_f(u^c)\,g^{ad} - \nabla_e(u^c)\,\nabla_f\left(u^b\right)\,g^{ad} - \nabla_e\left(u^d\right)\,\nabla_f(u^a)\,g^{bc} - \nabla_f(\nabla_e(u^a))\,g^{bc}\,u^d \\ &- \nabla_f\left(\nabla_e\left(u^b\right)\right)\,g^{ad}\,u^c - \nabla_f(\nabla_e(u^c))\,g^{ad}\,u^b - \nabla_f\left(\nabla_e\left(u^d\right)\right)\,u^a\,g^{bc}\Big)\end{aligned}$$

Motivo: La expresión se conserva simbólica: su proyección con el campo genérico Phi(r, varphi) contiene 32 nodos de derivada covariante, por encima del límite seguro 24 del backend. Especialice el perfil escalar o eleve el límite del backend de componentes. La corrida no solicitó validación Wolfram/xAct.

Extras: perfiles escalares de una variable, con la misma métrica del ansatz genérico. Se sustituyen únicamente las componentes de L y P^{abcd} ya calculadas; sin validación xAct independiente de estos perfiles. Las componentes completas de estos extras constan en `presentation.json`.

Extra: $\phi = \Phi(r)$

$$L = -\frac{1}{2\,r}\,\left(\left(\alpha\,\Phi'(r)^2\,f(r)+4\right)\,f'(r)+r\,\left(\alpha\,\Phi'(r)^2\,f(r)+2\right)\,f''(r)\right)$$

Componentes independientes no nulas; $P^{abcd} = -P^{bacd} = -P^{abdc} = P^{cdab}$.

$$\left[P^{abcd}\right]^{0101} = -\frac{\alpha\,f(r)\;\Phi'(r)^2}{4} - \frac{1}{2}$$

$$\left[P^{abcd}\right]^{0202} = -\frac{1}{2\,r^2\,f(r)}$$

$$\left[P^{abcd}\right]^{1212} = \frac{\left(\alpha\,\Phi'(r)^2\,f(r)+2\right)\,f(r)}{4\,r^2}$$

Extra: $\phi = \Phi(\varphi)$

$$L = -\left(\frac{2\,f'(r)}{r} + \frac{\alpha\,\Phi'(\varphi)^2\,f'(r)}{r^3} + f''(r)\right)$$

Componentes independientes no nulas; $P^{abcd} = -P^{bacd} = -P^{abdc} = P^{cdab}$.

$$\left[P^{abcd}\right]^{0101} = -\frac{1}{2}$$

$$\left[P^{abcd}\right]^{0202} = -\frac{\left(2\,r^2+\alpha\,\Phi'(\varphi)^2\right)}{4\,r^4\,f(r)}$$

$$\left[P^{abcd}\right]^{1212} = \frac{\left(2\,r^2+\alpha\,\Phi'(\varphi)^2\right)\,f(r)}{4\,r^4}$$

Política de lectura. Este PDF/LaTeX es una vista. El archivo `results.json` conserva la representación intermedia canónica y la trazabilidad completa.